Egg Segmentation and Individual Tracking

Status: COMPLETE_LOCAL_RESULTS | **Generated UTC:** 2026-10-31T13:51:34.283868+00:00

Reviewer quick path: read this report, inspect the linked annotated video and view the saved metrics/overlays. Re-annotating 110 images is not required to review the submission. Full Colab reproduction instructions are in README.md and report/REPRODUCE.md.

Open public annotated-video output

Open public GitHub repository

AI-use disclosure: Arena.ai assisted planning, draft visual grouping, code and documentation. The user executed the pipeline, selected/reviewed prompts and remains responsible for verification. No unexecuted result is claimed.

Assignment requirement-to-evidence map

Requirement	Evidence in this submission
Task 1 — pretrained segmentation	Steps 05–06; pretrained SAM ViT-B; 110 reviewed references; prompts, masks and review status.
Task 2 — 4:1 split and classical trials	Steps 03–04 and 07; frozen 88/22 group split; all 16 DEV configurations and selection log.
Task 3 — quantitative and qualitative evaluation	Step 08; frozen EVAL metrics, per-image counts, worst-case table and actual comparison panels.
Task 4 — individual video tracking	Step 09; persistent raw/boiled IDs, full-video CSV/JSON and public corrected Drive MP4.
Submission and reproduction	This PDF in GitHub; clickable public links; README, REPRODUCE, requirements and scripts 01–10.

1. Task and dataset

The supplied Roboflow v1 classification dataset contains 110 original 640 x 640 JPG images but no independent pixel-level gold masks. The work covers pretrained-SAM reference creation, a separate classical segmentation method, group-aware development/evaluation and individual egg tracking on the full supplied video.

Reference limitation: the masks are human-reviewed SAM-derived silver references. Classical scores are agreement with these references, not absolute accuracy against independent human ground truth.

2. Leakage-reduction split

Related visual setups were grouped before experiments and frozen into 88 development and 22 held-out evaluation images. Declared group crossings: 0. Evaluation groups: G06, G07, G08, G09, G10, G12, G13.

The conservative 58-image kitchen family G11, including the clearly visible multi-egg scenes, remains development-only. Held-out multi-egg and scene diversity are therefore limited. Visual families reduce obvious setup leakage but cannot prove unique people, physical eggs or capture sessions without provenance.

3. Pretrained SAM reference creation

Pretrained SAM ViT-B was used without training or fine-tuning. Official commit: dca509fe793f601edb92606367a655c15ac00fdf. Reviewed references: 110/110. Automatic grid candidates or manual boxes supported prompt discovery; accepted instances record point+box or human-box prompts and human review decisions.

Checkpoint SHA-256: ec2df62732614e57411cdcf32a23ffdf28910380d03139ee0f4fcbe91eb8c912

The label policy marks visible egg pixels only, excludes hand/background and does not invent hidden regions. Empty images use explicitly confirmed empty references. Binary union, instance masks, prompts and review status are retained.

4. Classical development experiments

Classical inference receives only original pixels and a configuration, never SAM/reference prompts or boxes. 16 fixed configurations using HSV/Lab seeds, morphology, connected components and/or GrabCut were compared on DEV only.

Selection used maximum macro DEV mean IoU, then mean Dice and micro IoU tie-breaks.

Selected: **C08_grabcut_center**. The method was frozen before held-out reference pixels were read.


```
{
  "center_patch": 0.16,
  "center_weight": 1.0,
  "close_kernel": 5,
  "component_filter": true,
  "grabcut_iters": 4,
  "lab_scales": [
    45.0,
    22.0,
    22.0
  ],
  "lab_threshold": 1.2,
  "max_area_frac": 0.65,
  "max_components": 2,
  "max_side": 384,
  "method": "grabcut_rect",
  "min_area_frac": 0.001,
  "open_kernel": 3,
  "rect_scale": 0.5,
  "sat_max": 80,
  "sure_bg_kernel": 15,
  "sure_fg_kernel": 9,
  "val_min": 70,
  "warm": false,
  "warm_hue_max": 35,
  "warm_hue_min": 3,
  "warm_sat_max": 180
}
```


5. Segmentation results

Split / aggregation	IoU	Dice	Precision	Recall
DEV macro	0.5107	0.5975	0.6367	0.6182
EVAL macro	0.6177	0.7346	0.7487	0.8288
EVAL micro	0.5928	0.7443	0.7384	0.7503

Definitions: IoU = TP/(TP+FP+FN); Dice = 2TP/(2TP+FP+FN); precision = TP/(TP+FP); recall = TP/(TP+FN). Macro averages per-image values with equal image weight. Micro pools pixel counts before calculating the metric. These are overlap measures, not classification accuracy.

EVAL median IoU: **0.6955**. Empty policy: both prediction and reference empty scores 1; other zero-division cases score 0. Per-image TP/FP/FN/TN, masks and overlays are saved. EVAL was not used for parameter tuning.

Development trial log

Config	Method	IoU	Dice	Precision	Recall	Errors
C08_grabcut_center	grabcut_rect	0.5107	0.5975	0.6367	0.6182	0
C09_grabcut_wide	grabcut_rect	0.4950	0.5976	0.5217	0.7990	0
C16_grabcut_slow	grabcut_rect	0.4950	0.5961	0.5225	0.7936	0
C10_grabcut_full	grabcut_rect	0.4678	0.5786	0.4755	0.8928	0
C01_hsv_pale	hsv	0.2572	0.3389	0.2658	0.6170	0
C15_lab_single	lab_center	0.2302	0.2967	0.2817	0.4366	0
C05_lab_tight	lab_center	0.2221	0.3089	0.2943	0.4416	0
C12_hybrid_lab	hybrid_lab	0.2079	0.2659	0.2248	0.4258	0
C06_lab_medium	lab_center	0.2035	0.2710	0.2294	0.4461	0
C14_hsv_single	hsv	0.1603	0.1902	0.1634	0.2895	0

Config	Method	IoU	Dice	Precision	Recall	Errors
C07_lab_broad	lab_center	0.1382	0.1865	0.1463	0.3337	0
C11_hybrid_hsv	hybrid_hsv	0.1354	0.1723	0.1362	0.3040	0
C02_hsv_medium	hsv	0.1148	0.1550	0.1161	0.2895	0
C13_hybrid_warm	hybrid_hsv	0.0802	0.1041	0.0803	0.2040	0
C04_hsv_warm	hsv	0.0582	0.0824	0.0583	0.1803	0
C03_hsv_broad	hsv	0.0526	0.0729	0.0527	0.1585	0

The grid compared HSV/Lab appearance seeds, warm-color inclusion, GrabCut rectangle sizes, hybrid refinement, component limits and iteration counts. Non-selected trials remain visible rather than being erased. Complete parameter JSON, micro metrics, runtime and per-image rows are retained in experiments/classical_dev_summary.csv and classical_dev_per_image.csv.

Lowest-IoU held-out cases

Image	Group	IoU	Dice	Pred. area	Ref. area
I019	G08	0.1792	0.3039	59629	10739
I033	G08	0.2015	0.3354	62993	19779
I017	G07	0.2808	0.4385	65897	20237
I034	G08	0.3623	0.5319	76737	32935
I038	G07	0.3850	0.5559	80006	130366
I027	G09	0.4303	0.6017	46434	20009
I030	G08	0.4378	0.6089	88334	154961
I101	G13	0.5067	0.6726	54500	27874
I104	G13	0.5369	0.6987	79614	49536
I016	G06	0.5653	0.7223	55639	96959

The following saved panels show classical prediction, reference and error pixels. Failure causes must be assigned only after visual inspection, not inferred from the score alone.

I019 - IoU 0.1792



I033 - IoU 0.2015



I017 - IoU 0.2808



6. Full-video individual tracking

Status: FULL_VIDEO_TRACKED. Frames processed: 1440 / 1440. Tracker: detection_gated_hsv. Persistent labels: boiled, raw.

Candidate segments: 2; not-detected object-frames: 205; out-of-scene object-frames: 870. Detection-gated tracking uses HSV/contour candidates, shape/size gates, one-to-one temporal association and green-card suppression. not_detected and out_of_scene rows have blank coordinates; boxes are hidden rather than drifted or interpolated. OpenCV visualization has no audio. Execution counts alone do not establish visual correctness or measured tracking accuracy.

Per-object status accounting: boiled: tracked=716, initialized=0, not_detected=68, out_of_scene=435; raw: tracked=647, initialized=0, not_detected=137, out_of_scene=435. Status counts describe visibility handling; no frame-by-frame tracking ground truth or accuracy percentage is claimed.

Selected annotated file: video_tracked_detection_v2.mp4; verified frames: 1440; FPS: 24.0; resolution: 1920 x 1080. SHA-256: 2716d27c26ac92168cbf7f4b10b1ccef307a04d0234e705eca73832c673c3689. The Drive URL must open this same accepted file.

Cut detection and tracking may fail under occlusion, abrupt motion or ambiguous identity. The complete rendered video must be visually reviewed and observed issues disclosed. Raw/boiled labels come from the video sequence; the tracker is not a learned raw/boiled classifier.

Open the publicly accessible Drive tracking video

7. Reproducibility and repository setup

Recommended reproduction environment: Google Colab. Use a GPU for full SAM candidate generation, or the documented CPU box-only fallback. Classical experiments, final evaluation, tracking and packaging run on CPU. Reviewers can inspect saved references and results without repeating the 110-image manual review.

```
from google.colab import drive
drive.mount('/content/drive')
%cd /content/drive/MyDrive/Egg_CV_Task
%run scripts/01_prepare_manifest.py
%run scripts/02_make_contact_sheets.py
%run scripts/03_prepare_scene_groups.py
%run scripts/04_freeze_split.py
%run scripts/05_setup_sam.py
%run scripts/06_sam_widget.py
%run scripts/07_classical_dev_experiments.py
%run scripts/08_classical_final_eval.py
%run scripts/09_track_full_video.py --tracker detection
%run scripts/10_build_submission.py --final --video-link 'https://drive.google.com/file/d/1dbA3VdnHSSiAlocSz3NPuhAUG_yzXlcC/view?usp=
```

The original dataset and authorized video must be placed at the documented project paths. Raw images, the 375 MB checkpoint and large videos are excluded from GitHub; relative paths and hashes are recorded. Exact package/GPU information is in metadata/sam_environment.json.

8. Limitations and integrity

1. SAM-derived references are not independent gold and may contain review errors.
2. Visual grouping is conservative and cannot guarantee session or physical-egg independence.
3. Whole-group splitting limits held-out scene and multi-egg diversity.
4. DEV model selection can overfit; held-out results were not used for further tuning.
5. Video tracking uses manual segment-start boxes and explicit loss records, not a learned raw/boiled classifier.
6. AI assistance is disclosed; code, annotations and outputs require human verification.

9. Public submission links

Public annotated-video output

Public GitHub repository

Public accessibility must be tested in an incognito/logged-out browser before submission. The annotated MP4 is intentionally kept on Drive rather than duplicated in GitHub because the assignment explicitly requires a public Drive link in this PDF. See report/GITHUB_PUSH.md and report/REPRODUCE.md for push and experiment instructions.